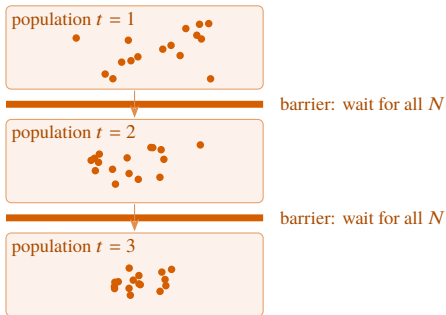
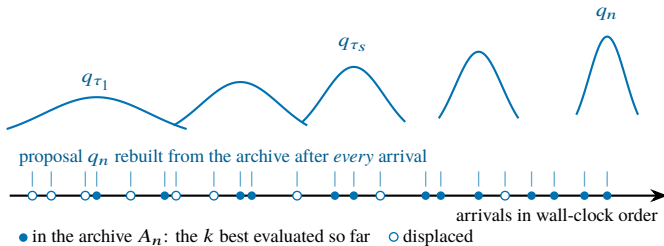


Generation-staged ABC-SMC / PMC



next proposal from the whole previous population;
each stage weighted against one previous proposal

Generation-free (ours): stage size one



Every evaluated particle is reweighted against the mixture of the proposals actually used (streaming AMIS, balance heuristic):

$$W_{i,n} = \pi(\theta_i) K_{\epsilon_n}(\rho_i) / \bar{q}_n(\theta_i), \quad \bar{q}_n = \text{prior floor} + \sum_s \alpha_s q_{\tau_s}$$

The reported posterior is a pure function of the evaluated history: replay the log and it comes back identical.